

Electric Vehicle Charge and Range Analysis

By: Nammi Manasa

Data Preprocessing Steps

1. Collected EV datasets

- Imported multiple CSV datasets containing EV specifications, charging stations, pricing, and vehicle information.

2. Removed duplicate records

- Identified and deleted duplicate rows to ensure data consistency.

3. Handled missing values

- Replaced missing categorical values with "**Unknown**" where appropriate.
- Removed rows with excessive missing data when required.

4. Removed unwanted symbols

- Removed symbols such as **€, £, km, kWh, sec, and Lakh** from columns to enable numerical analysis.

5. Converted data types

- Converted text-based numeric columns (Price, Range, Battery Capacity, Efficiency, Fast Charge Speed, etc.) into numeric data types.

6. Created calculated fields

- Developed calculated measures and KPIs such as:
 - Average Price
 - Average Range
 - Average Efficiency
 - Total EV Models
 - Total Charging Stations

7. Filtered invalid records

- Removed null or irrelevant values from visualizations and filters to improve dashboard clarity.

8. Validated data quality

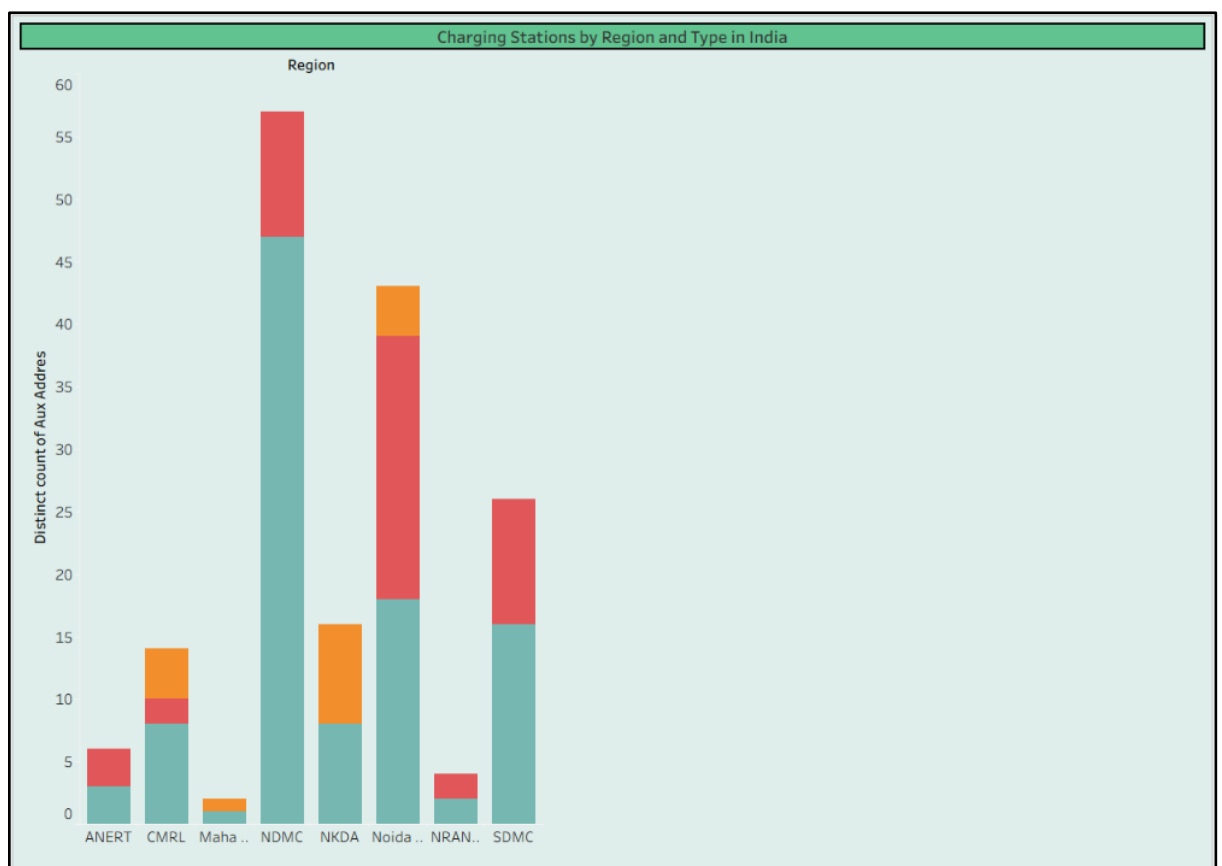
- Verified that all fields had correct data types and values before creating charts and dashboards.

9. Prepared data for visualization

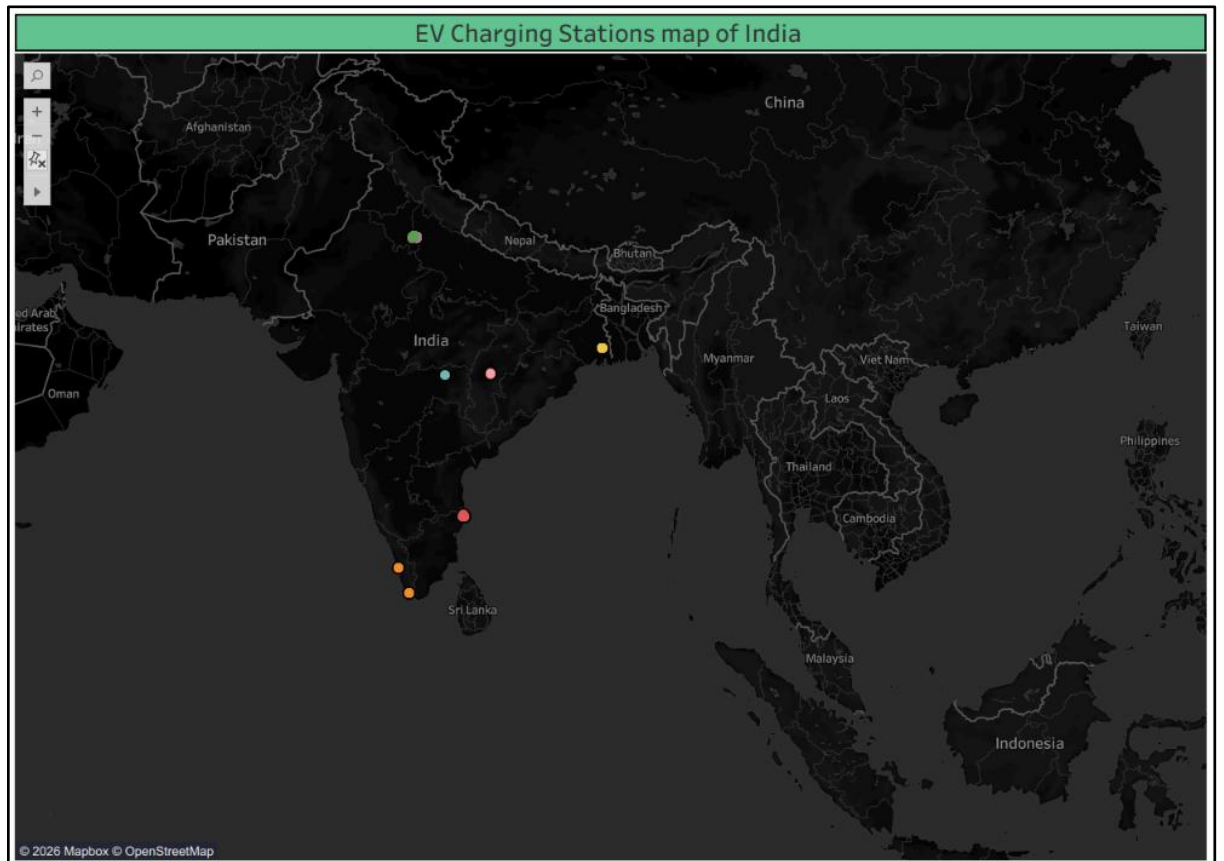
- Loaded the cleaned datasets into Tableau Desktop and used them to build dashboards, stories, and interactive visualizations.

Business Questions

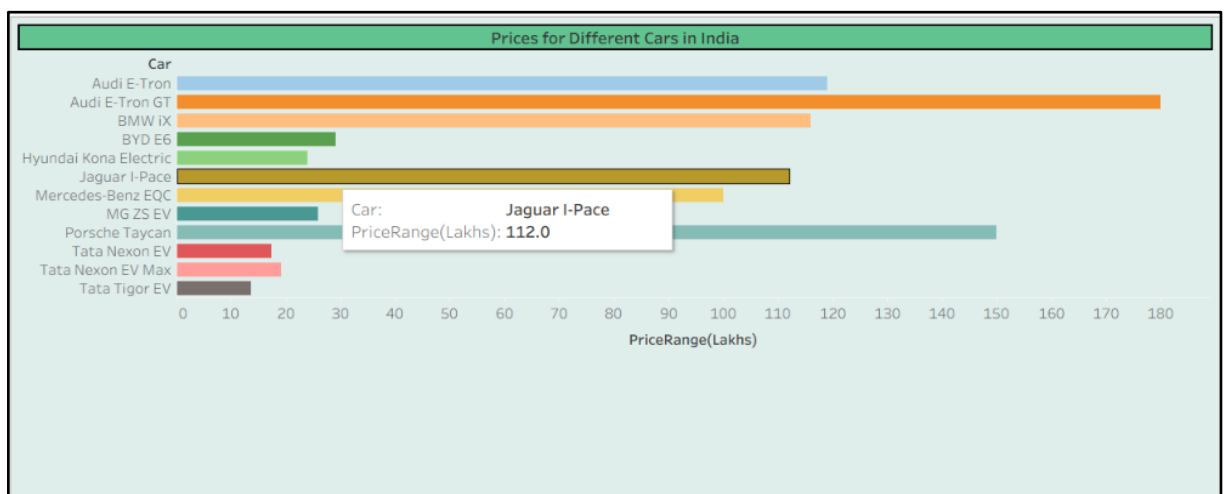
1. Which regions in India have the highest number of EV charging stations??



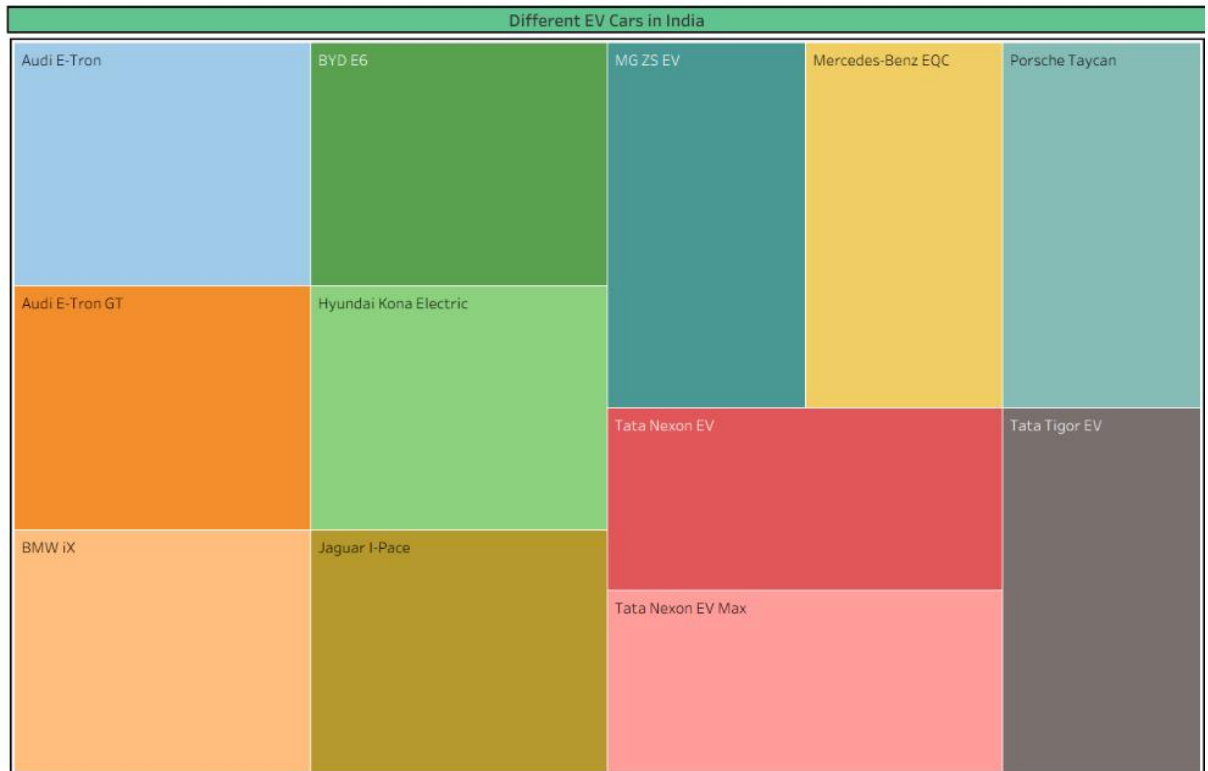
2. How are EV charging stations geographically distributed across different regions of India?



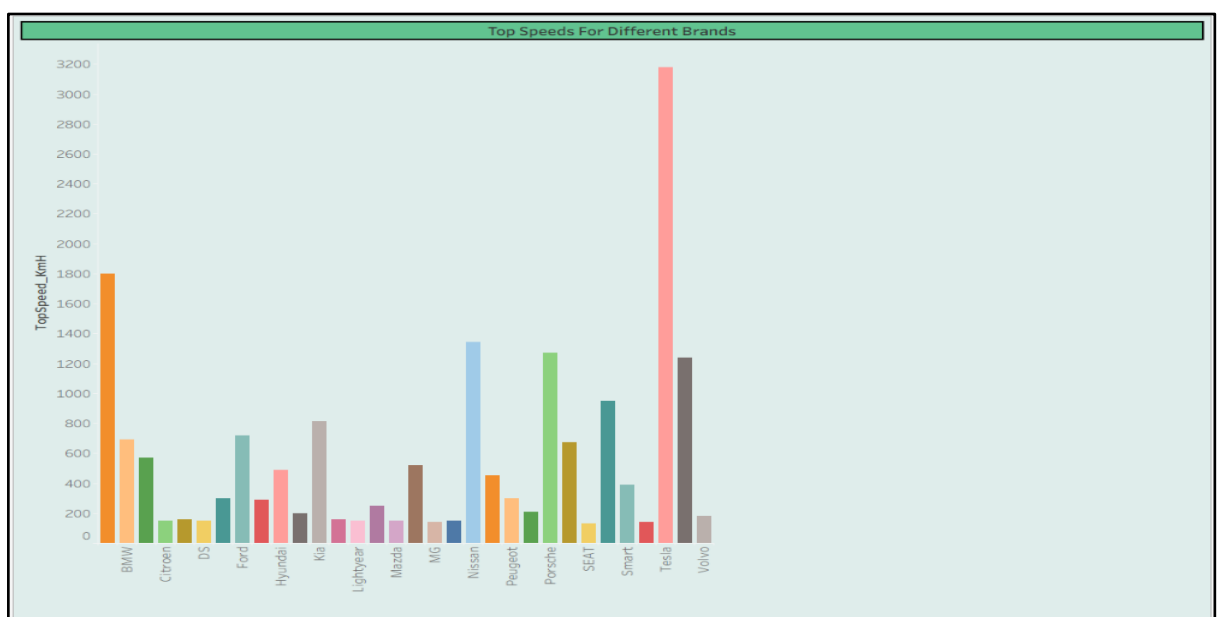
3. Which electric vehicle models are the most expensive, and how do their prices compare?



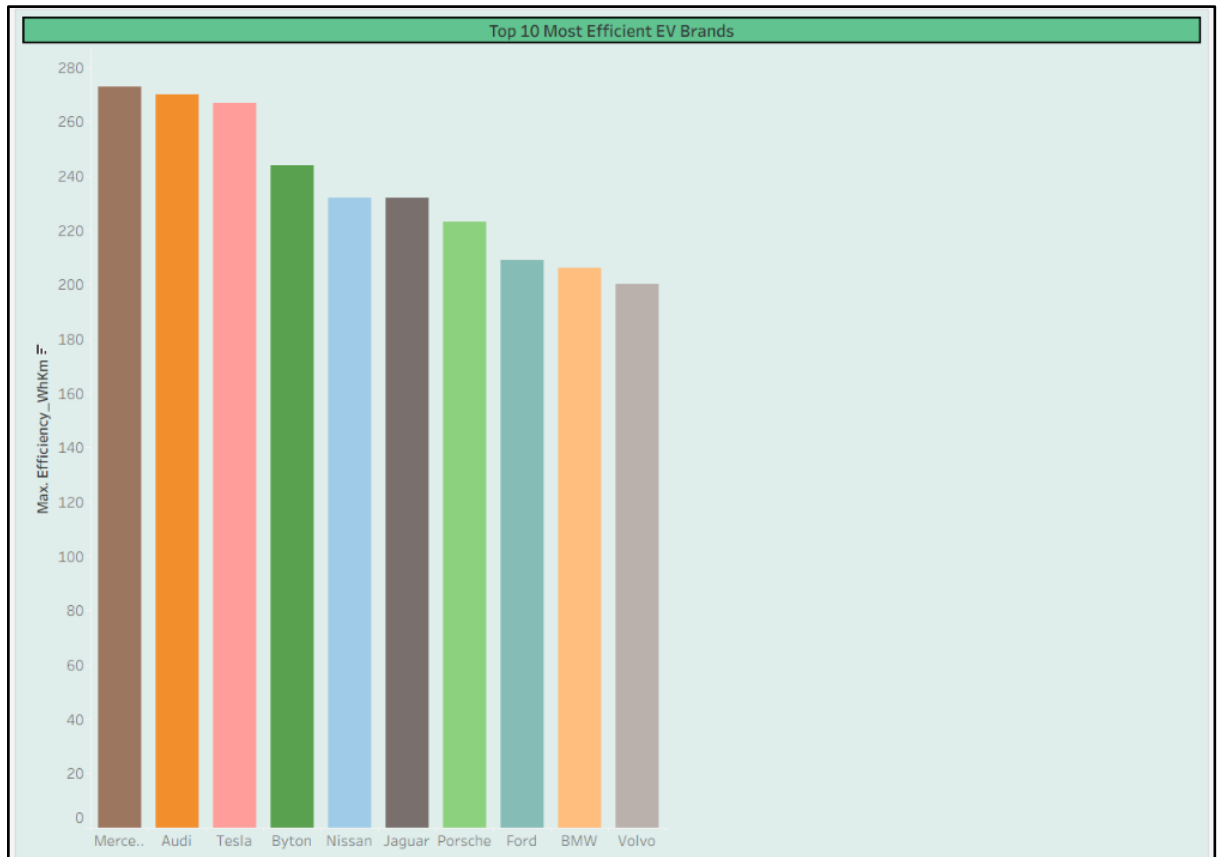
4. Which EV models occupy the largest share among the selected electric vehicles in the dataset??



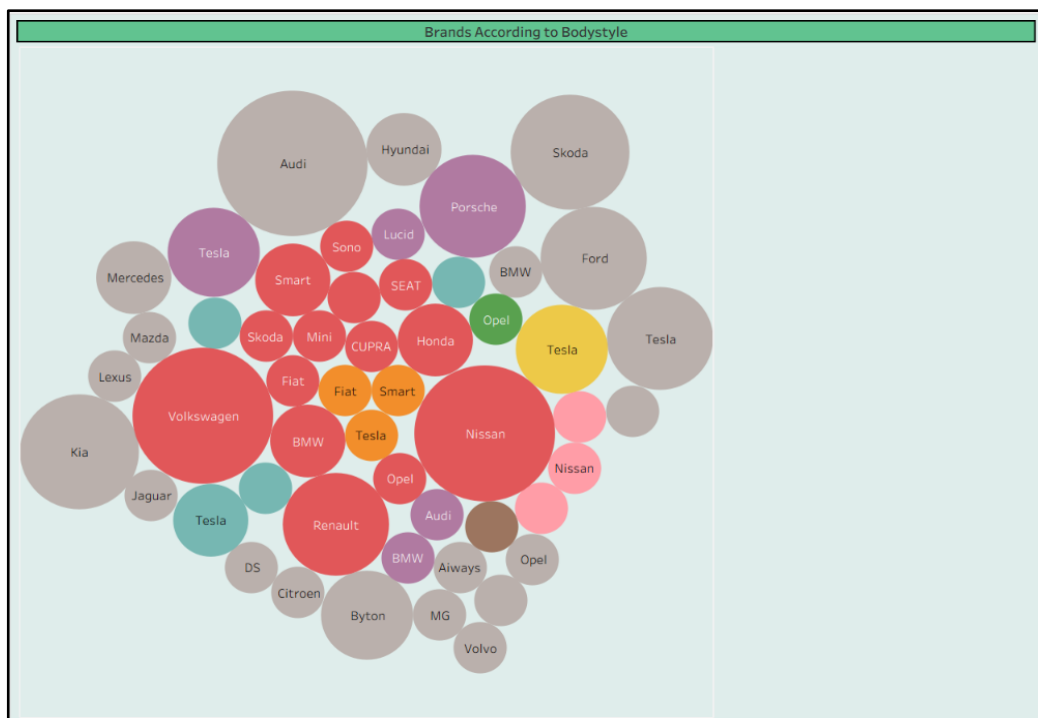
5. Which electric vehicle brands offer the highest top speed?



6. Which electric vehicle brands provide the highest energy efficiency (Wh/km)?



7. How are different EV brands distributed across various body styles?



8. How are electric vehicle brands distributed across different powertrain types?

